nawal@guerd:~$ whoami

Nawal Guerd

Open to DevOps, cloud and platform roles

DevOps & Cloud Infrastructure Engineer

Kubernetes · CI/CD & GitOps · Infrastructure as Code · Automation · Cloud-native · Security by design

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github.com/louane20 louane20.github.io/portfolio

2 years running production Kubernetes at a national mobile operator

Multi-DC bare-metal platforms, including air-gapped sites

IEEE WCNC 2025 co-author, Zero Trust for cloud-native 5G

3 working languages, Arabic, French and English

PROFILE

DevOps engineer who loves **cloud-native infrastructure and automation**. Since November 2024 I have built and run **production Kubernetes** platforms for Algeria's national mobile operator: bare-metal clusters across several datacenters, some of them air-gapped, delivered with **GitLab CI/CD and Argo CD**, automated with **Ansible** and secured with Cilium and Vault. A security background (M.Sc., IEEE WCNC 2025 paper on Zero Trust) means I design infrastructure that is secure from the first commit. Looking for a DevOps, cloud or platform engineering role where automation and reliability matter.

SKILLS

`highlighted` : in production

Kubernetes and cloud-native platforms	**Kubernetes**, **kubeadm**, containerd, Docker, Docker Compose, Linux
Automation and infrastructure as code	**Ansible**, **HashiCorp Vault**, Python, Shell scripting
CI/CD and GitOps	**GitLab CI/CD**, **Argo CD**, GitHub Actions, Trivy, cosign
Networking and high availability	**MetalLB**, **F5**, **Kong**, **Nginx**
Observability and data services	**Prometheus**, **Grafana**, **Loki**, **K6**, **Kafka (Strimzi)**, **PostgreSQL**, **Redis**, Promtail, cAdvisor, Node Exporter, ClickHouse, Oracle
Security and Zero Trust	**Cilium**, **Network policies**, Istio, Calico, Nmap, Wireshark, Snort, OpenVAS, OWASP ZAP
Languages	Python, Shell, Node.js, C, PHP
Systems	Linux, Windows Server, Active Directory
Cloud and telecom	Microsoft Azure, 5G Core, RAN, O-RAN, 3GPP

EXPERIENCE

DevOps Engineer · ATM Mobilis

Nov 2024 – Present · Algiers, Algeria

Algeria's national mobile network operator

Production · Bare metal · Air-gapped · Multi-datacenter

- Build and operate **production Kubernetes** clusters (kubeadm, bare metal) across **multiple datacenters**, including **air-gapped** sites reachable only through a DMZ and gateways.
- Automate the provisioning and configuration of servers and clusters with **Ansible**: code the whole team can rerun.
- Design **CI/CD and GitOps** delivery with **GitLab CI/CD** and **Argo CD**: every change is versioned in Git, reviewed and deployed through a controlled, auditable path.
- Secure the platform by design: **Zero Trust** network policies and segmentation with **Cilium**, secrets in **HashiCorp Vault**.
- Engineer traffic and **high availability**: exposure and load balancing with MetalLB, F5, Kong and Nginx, across datacenters.
- Run **observability** (Prometheus, Grafana, Loki), load-test with **K6**, and operate Kafka (Strimzi), PostgreSQL and Redis.
- Partner daily with development, security, network and infrastructure teams on troubleshooting and documentation.

Cybersecurity bootcamp · CyberTalents

Jul 2022 – Oct 2022 · Remote

- Trained in network security, penetration testing and vulnerability assessment.

Summer research intern · Sonatrach — Jul 2022 · Algiers, Algeria

- Assessed the IT infrastructure of several departments and proposed an open-source, containerized video-surveillance solution.

Summer intern · ManyDesigns Afrique — Jul 2020 – Aug 2020 · Algiers, Algeria

- Deployed enterprise web servers and designed a local network with its access-control policies.

SELECTED PROJECTS

This website, delivered like production software `louane20.github.io/portfolio`

The site, and the PDF résumé, are generated from a single content file and go through a full delivery pipeline on every change: automated tests, accessibility and performance budgets, security scans, a signed container image, and a Kubernetes test where Cilium must prove the network policies. Only a fully green run is deployed.

GitHub Actions · Playwright · Trivy · cosign · kind · Cilium

Automated deployment of a cloud-native 5G platform 2024 — Kubernetes · Istio · Automation · 5G

Master's thesis: an application that automates the deployment of a cloud-native 5G network on Kubernetes and the management of its clusters, with Zero Trust policies enforced by Istio. It led to the IEEE WCNC 2025 paper.

Securing microservices on Azure 2024 — Azure · Zero Trust · DevSecOps

Compared Zero Trust, perimeter-based and DevSecOps approaches for microservices, then built a security framework for a microservices deployment on Microsoft Azure.

DDoS detection and mitigation in SDN 2024 — Python · Mininet · Ryu · Machine learning

A software-defined network built with Mininet and the Ryu controller, with a Random Forest classifier that detects DDoS traffic and an automated, real-time mitigation module.

EDUCATION

USTHB: University of Science and Technology Houari Boumediene, Algiers

Master's degree, Computer Systems Security · USTHB — 2022 – 2024

Thesis: *Design and Implementation of a Zero Trust Security Solution for 5G Networks*, which led to the IEEE WCNC 2025 paper.

Bachelor's degree, Mathematics and Computer Science · USTHB — 2019 – 2022

Telecommunications and networks track, **ranked 2nd of 69**. Capstone: containerized a video-surveillance application with Docker Compose, with Prometheus and Grafana monitoring.

RESEARCH

A Multi-Layered Zero Trust Microsegmentation Solution for Cloud-Native 5G & Beyond Networks

IEEE WCNC 2025 · Milan, Italy, March 2025 · Chafika Benzaïd, **Nawal Guerd**, Nour El Houda Rehouma, Khaled Zeraoulia, Tarik Taleb

Proposes **THAALOUB**, a 3GPP-compliant Zero Trust framework for end-to-end security in cloud-native 5G and beyond networks.

`doi.org/10.1109/WCNC61545.2025.10978671` · `zenodo.org/records/14699793`

CERTIFICATIONS

Certified Kubernetes Administrator (preparation course)
KodeKloud, Aug 2025

GitLab CI Fundamentals
GitLab, Nov 2024

Next-Generation Cyber Security
Huawei, Aug 2024

Cybersecurity Essentials
Cisco, Aug 2023

Fundamentals of Deep Learning
NVIDIA, Dec 2022

AWARDS

Young Entrepreneur League 2023, finalist
Injaz El Djazair, 2023

HackINI CTF, 12th edition
Shellmate, ESI, 2023

JunctionX Algiers hackathon
MOBORN Tourism, 2021

LANGUAGES

Arabic · native

French · fluent

English · proficient

OPEN SOURCE AND COMMUNITY

IT team leader of the Openminds open-source club at USTHB (2020–2023). Organized Install Party 10 and Hacktoberfest, and ran workshops on open-source tools.

Capture-the-flag competitions, technical writing and research conferences.